



positive

negative





IPMs: Intuition

$$d_{\mathcal{F}}(\alpha, \beta) = \sup_{f \in \mathcal{F}} \int_{\mathcal{X}} f(x) d\alpha(x) - \int_{\mathcal{X}} f(x) d\beta(x)$$

- IPMs quantify distance through a **witness that 'exposes' differences** in probability
- They seek the **best witness** (in $\mathcal{F}$): one that maximally amplifies differences
- **Intuitively:**
 - If $d\alpha(x) \gg d\beta(x)$, f will put large **positive** value on x , so that $f(x)(d\alpha(x) - d\beta(x))$ is large/positive
 - If $d\alpha(x) \ll d\beta(x)$, f will put large **negative** value on x , so that $f(x)(d\alpha(x) - d\beta(x))$ is large/positive
 - $d\alpha(x) \approx d\beta(x)$, $f(x)$ can do whatever

Examples of IPMs